Introduction To Algorithms

Week 6 – **Graph Traversal and Modeling**

"Modeling Connected Things"

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Terms to Remember

Term	General Meaning	Graph / Network Meaning
Network	Any connected system	A graph model of connected nodes and edges
Node	A point or object	A specific item in the graph
Vertex	A corner or point	Another term for node
Edge	Outside boundary	A relationship or connection between nodes
Link	A connection	Another term for edge
Neighbor	Something nearby	A node directly connected to another node
Path	A route	A sequence of connected nodes
Traversal	Moving through something	Visiting graph nodes using an algorithm
Centrality	Importance	A metric estimating structural importance
Density	How compact something is	How many possible connections actually exist

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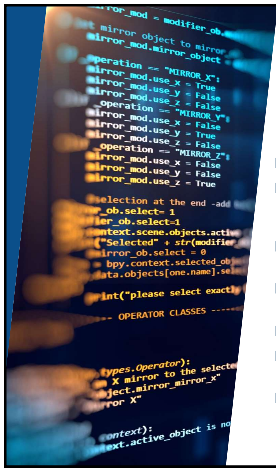

What We Will Use Today

- Graph vocabulary
- Directed, Undirected, Simple, and Weighted graph recognition
- Adjacency-list representation
- Neighborhoods and shortest-path intuition
- BFS and DFS traversal
- Queue and Stack state as evidence
- Model-limit explanation

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What We Will "Touch On" and Revisit Later

- similarity and recommendation
- ranking and centrality ideas
- AI/data relationship modeling
- explainability and limits of graph-based claims
- optional social network analysis context

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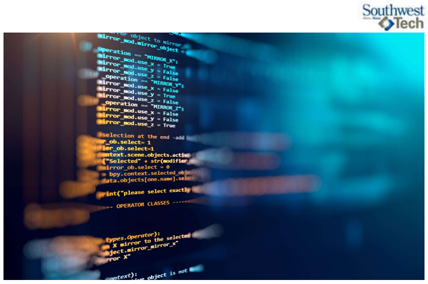

Review Coding Lab

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Review: What Lab 05 Taught Us

Last week - strategy choice depended on problem shape:

- correctness
- readability
- growth
- fit to data
- limitations

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Today's Question

How can we model connected parts so an algorithm can move through them?

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Graphs Model Relationships

A graph is a model of connected things.

It can represent:

- places connected by routes
- tasks connected by dependencies
- people connected by communication
- steps connected in a workflow
- rooms connected by paths

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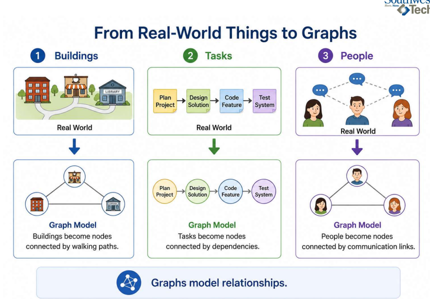

Graph Model

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What Counts as Success Today

- define nodes and edges
- build an adjacency list
- trace BFS or DFS
- explain traversal evidence
- name what the graph le

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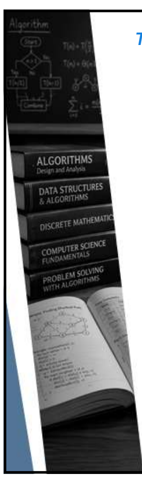

Textbook Review

Graph Algorithm Overview

The reading covers:

1. graph representation
2. network theory analysis
3. graph traversals
4. case study
5. neighborhood techniques

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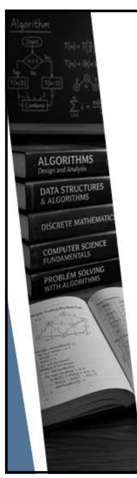

Textbook Review

Terms Become Precise

Some familiar words become technical:

- network
- node
- edge
- neighbor
- path
- traversal
- density
- Centrality

Use the graph meaning during this lesson

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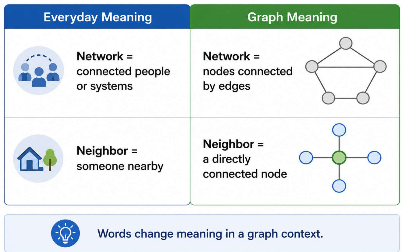

Term Precision

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Textbook Review

Do Not Panic Over Formulas

The reading includes advanced graph formulas.

For this week:

- notice the term
- read for the idea
- connect it to a simple graph
- do not try to master every formula

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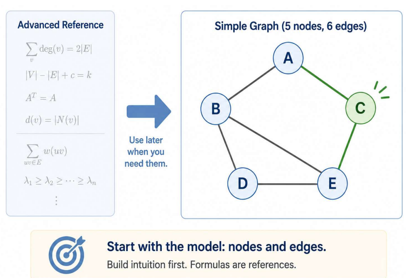

Formula Posture

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Textbook Review

Graph Basics

Core graph pieces:

- vertices or nodes
- edges or links
- network structure
- simple, directed, undirected, weighted types

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Textbook Review

Neighborhoods

A neighborhood asks:

- What is directly connected?
- What is one hop away?
- What is two hops away?
- What can be reached from here?

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Textbook Review

Traversals

The reading introduces:

- breadth-first search (BFS)
- depth-first search (DFS)
- traversal order
- specific searches using traversal

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Graph Basics and Representation

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Node Or Vertex

A node is one thing in the graph.

Examples:

- building
- person
- task
- course
- web page
- game room

Vertex is another term for *node*.

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Edge Or Link

An edge is a connection between nodes.

Examples:

- sidewalk between buildings
- prerequisite relationship
- communication link
- workflow transition
- door between rooms

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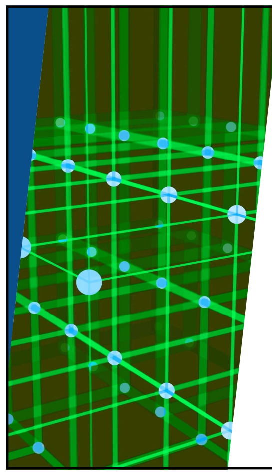

Network

In graph work, a network means:

- defined nodes
- defined edges
- a model of relationships

It does not mean "everything about the real system."

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Simple Graph

A simple graph usually avoids:

- duplicate edges
- self-loops
- extra edge complexity

It is often a good beginner model.

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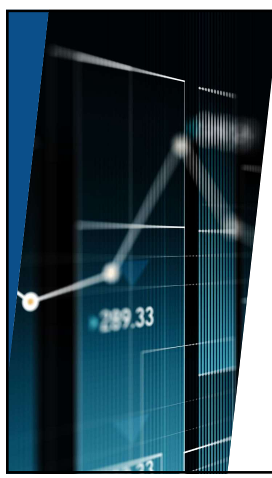

Directed Graph

A directed graph has one-way relationships.

Examples:

- course prerequisite
- workflow step
- escalation path
- one-way route

Direction changes what can be reached.

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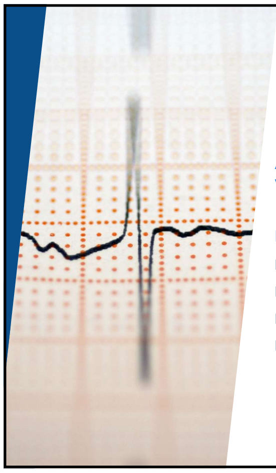

Undirected Graph

An undirected graph has two-way relationships.

Examples:

- sidewalk between buildings
- hallway between rooms
- mutual connection
- two-way transit route

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Weighted Graph

A weighted graph adds a value to edges.

Weights may represent:

- distance
- time
- cost
- risk
- strength of connection

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Adjacency List

An adjacency list stores each node with its neighbors.

Example:

Parking: Library, Student Center
Library: Parking, Science Hall, Cafeteria

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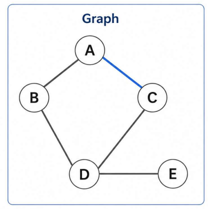
Graph



Adjacency List

A	B, C
B	A, D
C	A, D
D	B, C, E
E	D

Each list entry names direct neighbors.

Adjacency List

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Diagram And Adjacency List Must Agree

Check the model both ways:

- Does every diagram connection appear in the adjacency list?
- Does every adjacency-list connection appear in the diagram?
- Are directed edges shown clearly?
- Are two-way edges listed both ways?

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A Graph is a Model

A graph may leave out:

- distance
- time
- priority
- capacity
- direction
- human judgment
- real-world messiness

A model is useful, but incomplete.

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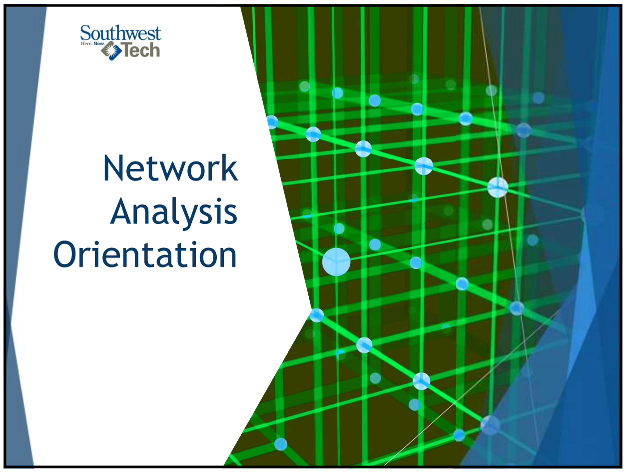

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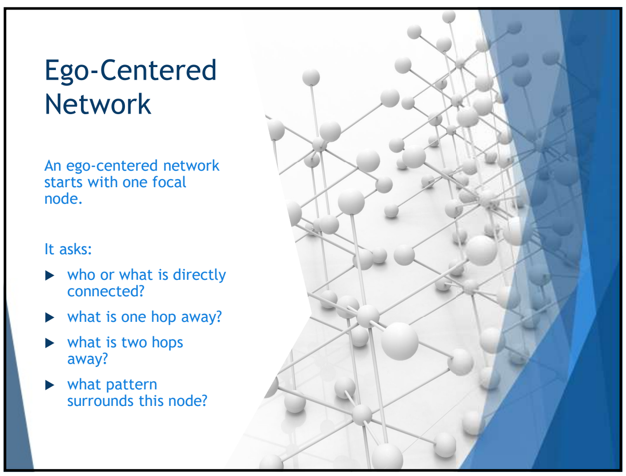

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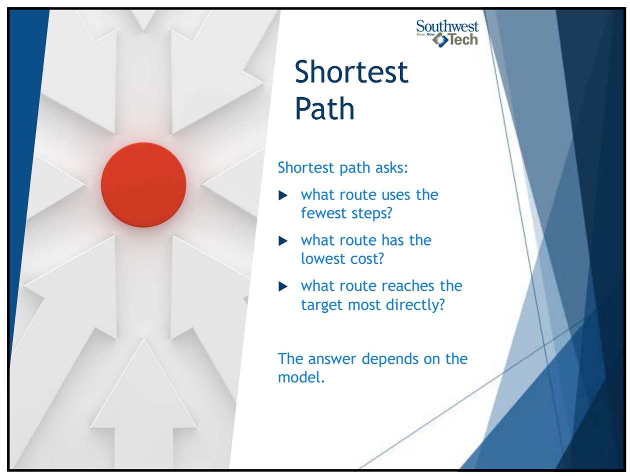

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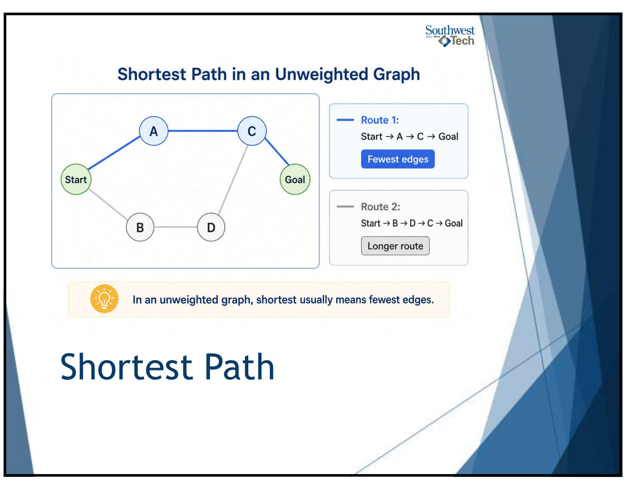

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Optional Metrics: *Recognition Only*

The optional reading includes:

- triangle
- density
- degree
- betweenness
- closeness
- eigenvector centrality

For this course: recognize the ideas.

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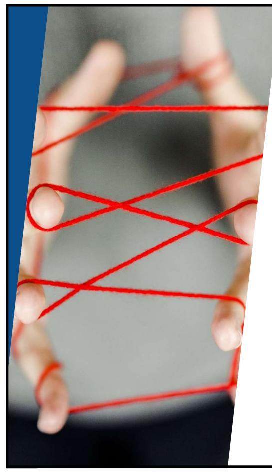

Degree And Density

Degree:

- How many edges connect to a node?

Density:

- How many possible connections actually exist?

Both describe structure.

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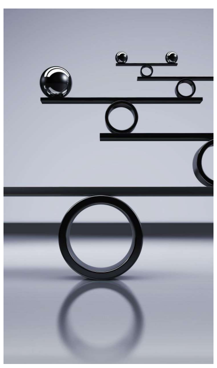

Centrality Is Not Importance By Itself

Centrality measures estimate structural position.

They do not automatically prove:

- value
- authority
- quality
- fairness
- truth

Metrics need interpretation.

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Break

10 Minute “Human” bio-break

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Graph Traversals

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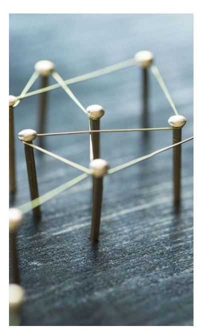

Traversal

Traversal means visiting nodes by following edges.

Traversal depends on:

- start node
- graph structure
- neighbor order
- traversal method

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Breadth-First Search

BFS explores outward by layers.

It tends to visit:

- the start node
- direct neighbors
- neighbors of neighbors
- later layers

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BFS Evidence

Step	Current Node	QUEUE State	Visited Nodes
1			
2 …			

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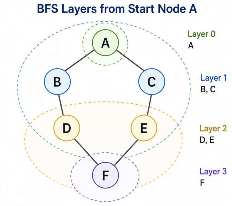
BFS Layers from Start Node A



Queue Evidence

Step	Visit	Queue
1	A	B, C
2	B	C, D
3	C	D, E
4	D	E, F

BFS explores nearest neighbors first.

BFS Evidence

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Depth-First Search

DFS follows a path deeply before backing up.

It tends to:

- choose a neighbor
- continue from that neighbor
- keep going until it cannot
- backtrack and try another path

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DFS Evidence

Step	Current Node	STACK State	Visited Nodes
1			
2 …			

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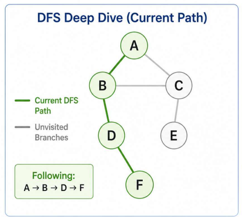
DFS Deep Dive (Current Path)



Stack Evidence

Step	Visit	Stack
1	A	C, B
2	B	C, D
3	D	C, F
4	F	C

DFS follows one path before backing up.

DFS Evidence

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BFS vs DFS

Question	BFS	DFS
Explores how?	outward by layers	deeply by path
Common state	queue	stack or call stack
Useful for	near/reachable layers	path exploration

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Neighbor Order Matters

Traversal order can change when:

- the start node changes
- the neighbor order changes
- the graph is directed
- the traversal method changes

The evidence should name the conditions.

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Optional Network Context

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Visualizing A Graph

A graph can be shown as:

- diagram
- adjacency list
- traversal table
- Python output
- library visualization

Each view reveals something different.

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Social Network Analysis Context

Social network analysis uses graphs to study relationships.

It may ask:

- who is connected?
- who bridges groups?
- where are clusters?
- how dense is the network?

Use caution when interpreting results.

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Fairness And Closeness

Graph metrics can influence decisions.

Ask:

- What data created the graph?
- What relationships were included?
- What relationships were ignored?
- Who could be misrepresented?

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What To Deeply Learn vs Recognize

Deeply learn now:
- nodes and edges
- adjacency list
- BFS and DFS
- traversal evidence
- model limits

Recognize for later:
- centrality metrics
- SNA
- graph libraries

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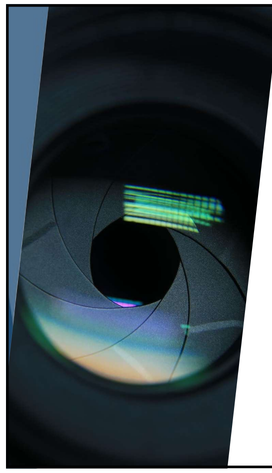

Demo

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Demo Scenario

Problem: Model a small campus route map with buildings as nodes and sidewalks as edges

Goal Demonstrate:
- a text diagram of the route map
- adjacency-list creation
- BFS traversal from one start building
- DFS traversal from the same start building

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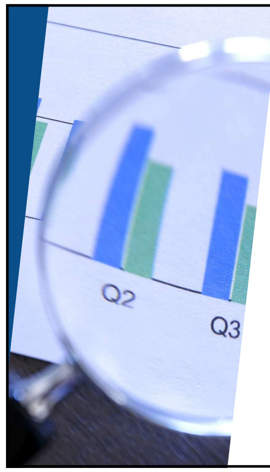

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Demo Evidence

- text diagram
- adjacency list
- BFS traversal table
- DFS traversal table
- traversal summary

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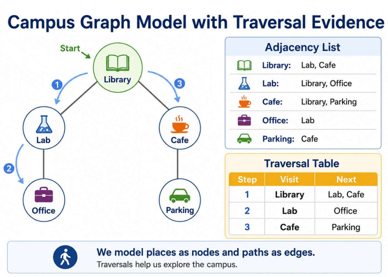

Demo Evidence

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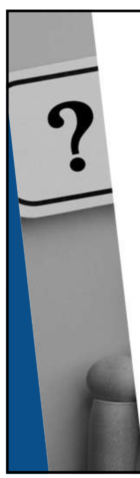

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Demo Key Questions

- What is the current node?
- What neighbors are available?
- What is waiting in the queue or stack?
- What has already been visited?
- What does the model leave out?

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Demo Explanation

"The graph model stayed the same, but BFS and DFS visited nodes in different orders because BFS explores outward by layers while DFS follows a path more deeply before backing up"

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From Demo to Lab

In Lab 06, you will model a different system.

Options include:

- workflow steps
- escalation paths
- transit stops
- game rooms
- prerequisites
- communication network
- grid movement map

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Lab 06 Evidence

- at least 6 nodes
- at least 7 edges
- adjacency list
- diagram or text representation
- BFS or DFS traversal
- traversal table
- model limitation

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README Evidence

Your README should show:

- What the graph models
- What the nodes represent
- What the edges represent
- What traversal you used
- What the traversal shows
- What the graph leaves out
- AI-use note, if applicable

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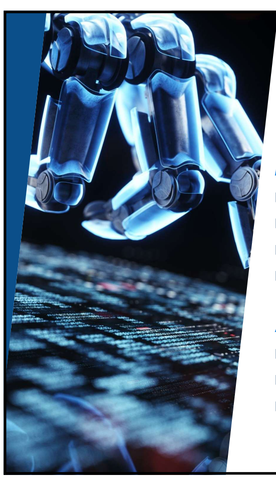

AI Use In Lab On

Manual first:

- draft your graph
- write your adjacency list
- choose your start node
- attempt traversal evidence

AI-assisted after:

- check diagram/list consistency
- explain BFS or DFS vocabulary
- critique model limits

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Wrap up

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What To Carry Forward

Graphs help us reason about connected systems.

Carry forward:

- Model the relationships
- Make connections explicit
- Trace the traversal
- Explain what the model shows
- State what the model leaves out

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Lab On Success Check

- Represents relationships clearly
- Uses graph vocabulary correctly
- Includes adjacency-list evidence
- Shows BFS or DFS behavior
- Explains model limits

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Next Step

1

For Lab 06:

- choose a system
- identify nodes and edges
- build an adjacency list
- create a diagram or text representation
- run or simulate BFS or DFS
- explain what the model leaves out

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How To Use The Textbook For Next Week's Reading

Next week connects algorithms to AI, data, recommendations, and security.

As you read:

- Focus on representation before advanced methods
- Notice how similarity and recommendation depend on chosen features
- Skim advanced ML language for recognition
- Learn the security vocabulary without trying to master cryptography
- Watch for limitations such as cold start, sparse data, and weakest link

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Thank you!

Have Fun!

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